Ziyu Peng

Boulder, Colorado

Undergraduate Student · Statistics & Data Science · University of Colorado Boulder

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EDUCATION

University of Colorado Boulder 2026–Present
B.S. in Statistics and Data Science, in progress

Xi'an Jiaotong-Liverpool University 2025–2026
First-year study in Data Science and Big Data Technology · **GPA 3.88/4.00**

RESEARCH & TECHNICAL PROJECTS

Equity Return Prediction and Low-Turnover Stock Selection 2026
2026 Shenzhen International FinTech Competition (FinTechathon) · Data Analytics Track

- Analyzed 7.9 million training observations from 4,650 A-share stocks; built 236 price-volume features and evaluated XGBoost and GRU models for cross-sectional return prediction.
- Developed separate stock-selection and daily-ranking layers. In 2022–2024 *development* validation, reduced Top-group Jaccard turnover from 0.76–0.80 to 0.02–0.04, with a trade-off in estimated excess return.

Ambiguity Is Not One Scalar 2026

- Implemented a visual question-answering router that enumerates plausible referents before choosing answer, clarify, or abstain.
- Evaluated a 26,425-example diagnostic benchmark; referent enumeration recovered 20 of 22 human-audited ambiguity misses versus 0 of 22 for an answer-consistency baseline. Code and results.

PW-OFPC: Object-Centric Predictive Coding 2025–2026

- Developed an abstract visual-reasoning pipeline that retains uncertainty over object attributes and compares candidates using precision-weighted residuals.
- Reported RAVEN macro accuracy of 0.857 for cross-option inference versus 0.537 for direct prompting; maintained scripts to reproduce archived evaluation tables. Code and results.

Corridor Wealth: Auditable AI Advisory Prototype 2026

- Built a synthetic-data financial advisory prototype with deterministic policy checks separate from model proposals and individually authorized simulated transactions. Project repository.

HONORS & RECOGNITION

FinTechathon 2026 — Data Analytics Track: Top 25 (top 10%), semifinal in progress; separately participating in the International AI Track.

CU Boulder Transfer Excellence Scholarship — $12,000 total (2026).

XJTLU Academic Scholarship (2026/27); **Third Prize**, XJTLU Biostatistics Competition (2026).

TECHNICAL SKILLS

Programming and tools: Python, R, PyTorch, XGBoost, Git/GitHub, Linux, LaTeX.

Methods: Feature engineering, statistical modeling, temporal validation, ablation studies, and reproducible evaluation.